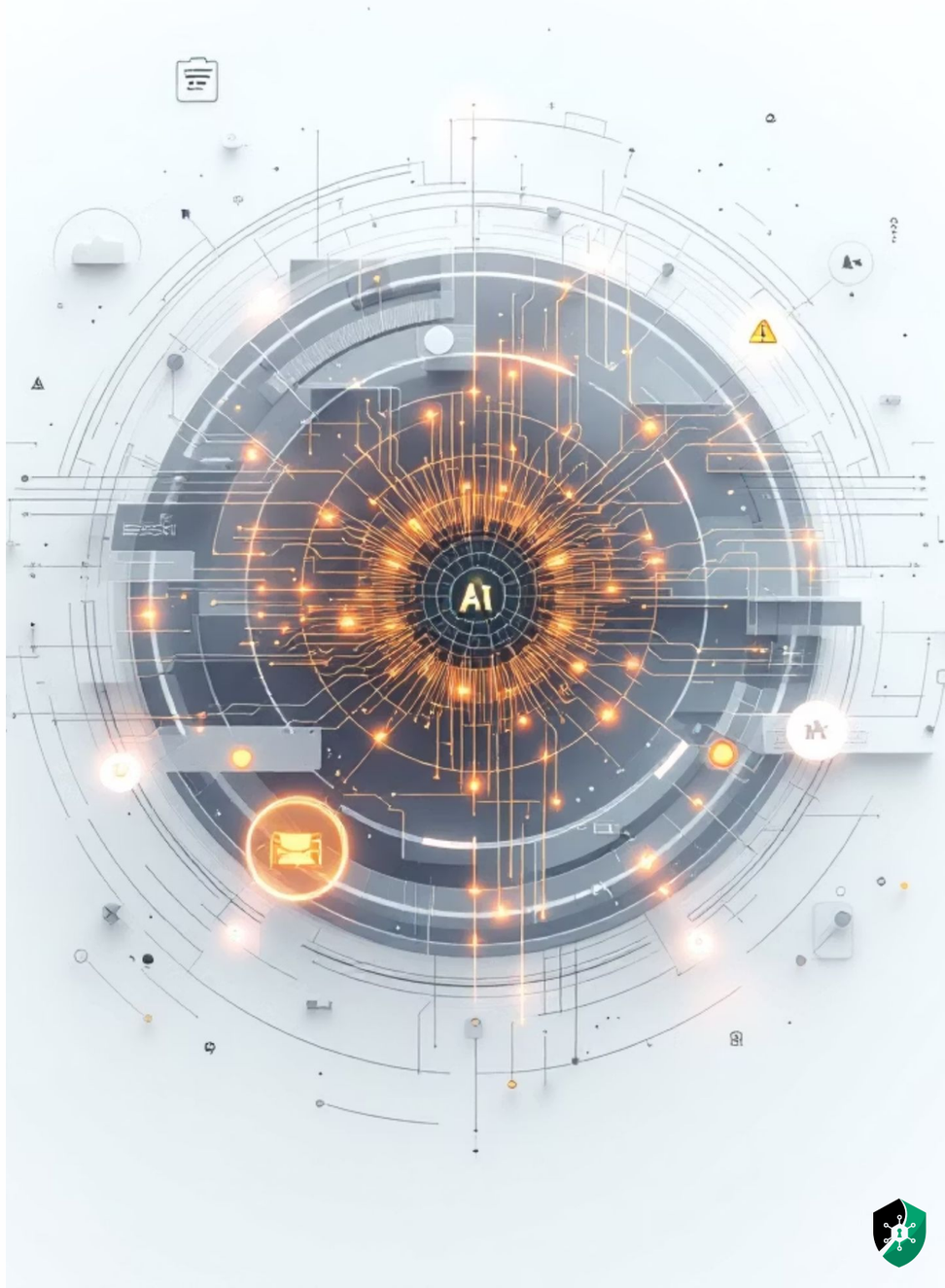
LESSON 6.2.2

Identifying Risks Unique to AI

AI introduces a distinct risk profile not just bugs, but learned biases, shifting realities, and adversarial threats. This lesson unpacks what makes AI risk different.



RISK #1

Biased or Unrepresentative Training Data

AI learns from data. If that data reflects historical inequities or omits key cases, the model inherits those flaws at scale.

🔍 Profile Your Data

Systematic audits to detect gaps, skews, and historical bias

🧪 Synthetic Data

Generate representative samples to fill underrepresented gaps

🌍 Real-World Eval

Evaluate model performance on actual production distributions

RISK #2

Distribution Shifts & Model Drift

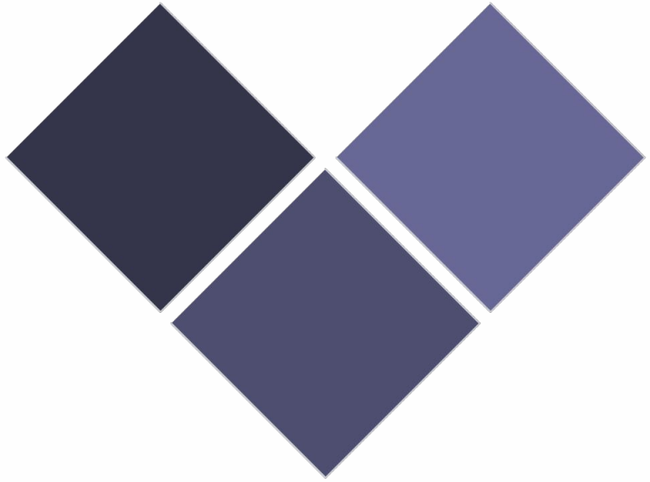
The world changes. When production data no longer matches training data, model accuracy quietly degrades often unnoticed.

Monitor

Compare live inputs/outputs to training baseline.

Rollback

Restore prior model version if retraining fails.



Retrain

Perform controlled retraining when drift is detected.

Proactive monitoring is the first line of defense. Rollback plans ensure continuity when drift is detected too late.

RISK #3

Interpretability Challenges

High accuracy ≠ transparency. Deep networks and foundation models can be right for the wrong reasons and we may never know why.

The Problem

- Complex models resist explanation
- Stakeholders can't audit decisions
- Regulatory environments demand accountability

Mitigations

- Feature importance analysis
- Comprehensive model documentation
- Prefer simpler models in regulated contexts

RISK #4

Adversarial & Privacy Attacks

AI systems face threats that go beyond traditional exploits — attackers can corrupt training data, reverse-engineer private records, or manipulate inputs to fool the model.

Data Poisoning

Modifies training data to steer model toward attacker-chosen behavior

Model Inversion

Reconstructs or detects private training records — a direct privacy risk

Adversarial Samples

Crafted inputs that cause deliberate misclassification

Prompt Injection

LLMs manipulated via untrusted content to leak data or change behavior

Defenses: Data integrity checks · Isolated training pipelines · Adversarial test sets · Rate limiting · Content filtering · Separate untrusted content from system instructions

RISK #5

Hallucinations & Unsafe Content

Models generate confident, fluent and sometimes entirely false outputs. Unchecked, this erodes trust and causes real harm.

Source Grounding

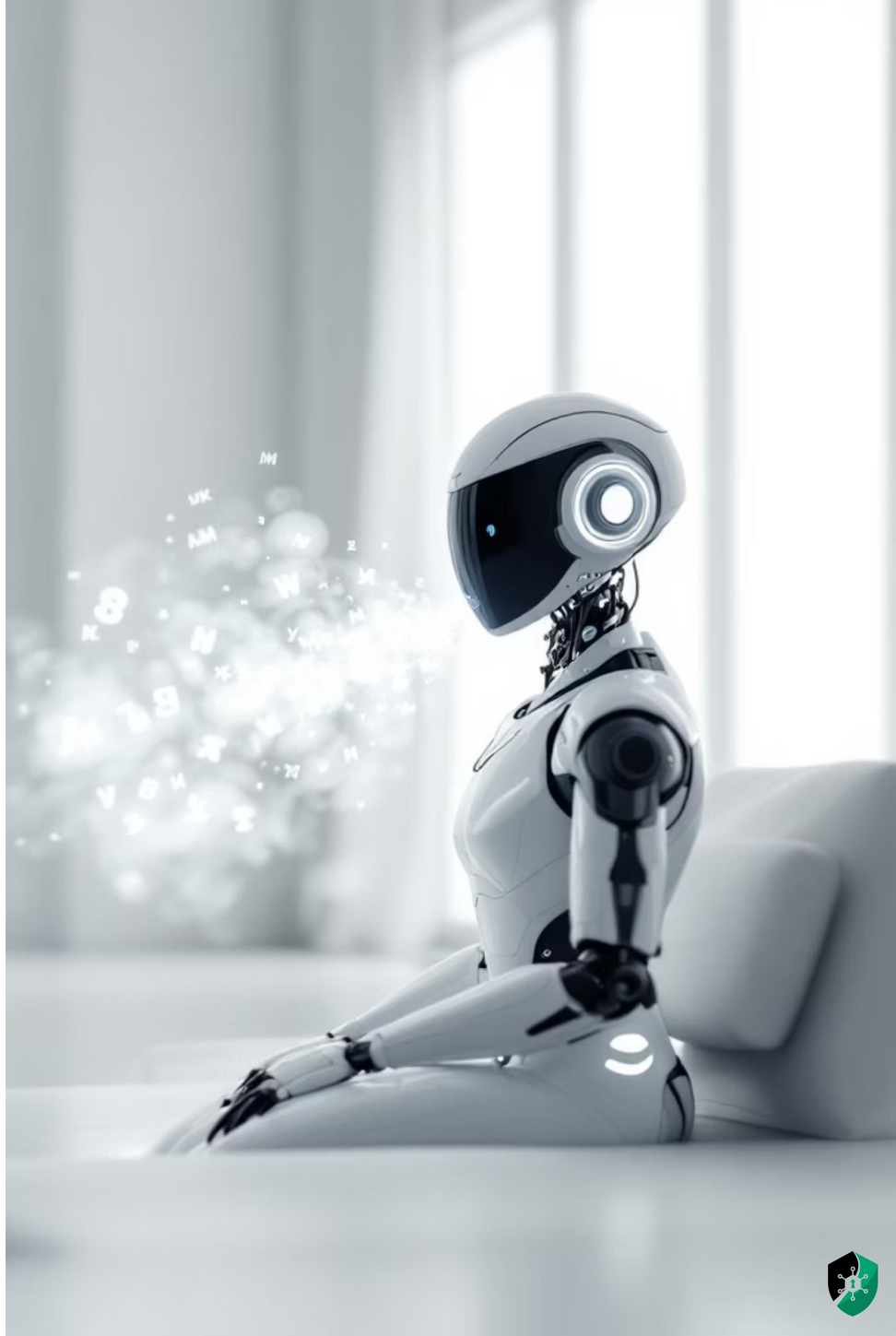
Bind outputs to verified sources using retrieval-augmented generation (RAG) with citations

Content Filters

Apply policy filters on both prompts and outputs to block toxic or harmful content

IP Leakage Control

Audit training data for proprietary material and unauthorized use creates legal and reputational risk



RISK #6

Third-Party & Supply Chain Risks

Most AI teams rely on external models, open weights, or community datasets. Silent updates can introduce new risks without warning.

Exposure Points

- External model providers updating without notice
- Open-weight models with unknown data provenance
- Community datasets with unverified quality

Governance Controls

- Supplier risk assessments
- Signed, versioned artifacts
- Internal validation before any upgrade

RISK #7

Over-Trust & Under-Trust

Human perception of AI is rarely calibrated. Both extremes — blind trust and unwarranted rejection, create operational risk.

⬆️ Over-Trust

Users accept confident AI outputs without scrutiny, bypassing critical judgment

⬇️ Under-Trust

Users dismiss valid AI outputs due to low awareness or past bad experiences

→ Training & Awareness

Educate users on AI capabilities and known failure modes

→ Interface Design

Surface uncertainty signals and supporting evidence in the UI

→ Procedural Safeguards

Mandate secondary human checks for high-impact AI-assisted decisions

SUMMARY

The AI Risk Landscape at a Glance

Biased Data

Historical inequities baked into training sets reproduce at scale

Distribution Drift

Real-world changes silently degrade model accuracy

Interpretability

Complex models resist explanation, limiting accountability

Adversarial Threats

Poisoning, inversion, injection — AI surfaces new attack vectors

Supply Chain

Third-party dependencies carry hidden versioning and provenance risks

LESSON 6.2.2

Key Takeaways

AI risk is data-driven

Biased or unrepresentative training data produces biased models. Systematic data profiling and real-world evaluation are non-negotiable.

Monitor for drift, plan for rollback

The world changes; training data doesn't. Continuous monitoring with rollback plans is essential for maintaining model reliability.

Interpretability enables accountability

When models can't explain themselves, governance fails. Prefer interpretable models in regulated settings; document all others rigorously.

Adversarial threats are AI-specific

Data poisoning, prompt injection, and model inversion require AI-native defenses: isolated pipelines, integrity checks, and content filtering.

Calibrate human trust deliberately

Design interfaces and training programs that help users neither over-trust nor under-trust AI outputs especially for high-stakes decisions.